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Section Editor, Journal of Animal Science
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Reducing crude protein and rumen degradable protein with a constant concentration of rumen undegradable protein in the diet of dairy cows: Production performance, nutrients digestibility, nitrogen efficiency, and blood metabolites

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Title Page

Running head: REDUCING CP AND RUP IN DAIRY COW

Reducing crude protein and rumen degradable protein with a constant concentration of rumen undegradable protein in the diet of dairy cows: Production performance, nutrients digestibility, nitrogen efficiency, and blood metabolites

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Conflict of Interest

The authors declare no conflict of interest.

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For Peer Review

ABSTRACT: The goals of ruminant protein nutrition are to provide adequate amounts of RDP for optimal ruminal efficiency and to obtain the desired animal productivity with a minimum amount of dietary CP. The aim of the present study was to examine effects of decreasing dietary protein by decreasing RDP with the optimum concentration of RUP on production performance, nutrient digestibility, N retention, rumen fermentation parameters, and blood metabolites in high-producing Holstein cows in early lactation. Nine multiparous lactating cows (2nd parities, averaging 50 ± 12 d in milk and milk yield of 48 ± 5 kg/d) were submitted to in a triplicate 3×3 Latin square design with 3 rations: 1) a TMR containing 16.4% CP (10.9% RDP based on DM), 2) a TMR containing 15.6% CP (10% RDP), and 3) a TMR containing 14.8% CP (9.3% RDP). The level of RUP was constant at 5.5% DM across the treatments. All the diets were calculated to supply a post-ruminal lysine to the methionine ratio of about 3:1. Dry matter intake, milk yield and composition, 4%FCM, and energy-corrected milk were not significantly affected with decreasing dietary CP and RDP levels. Cows fed 16.4% CP diets had greater ($P < 0.01$) CP and RDP intakes, which resulted in a trend toward greater concentrations of plasma urea N compared with other treatments. Daily N intake linearly decreased ($P < 0.01$) with decreasing dietary CP and RDP levels, whereas the intake of RUP and fecal N excretion (g/d) did not change. Apparent digestibility of nutrients, ruminal pH, and $\text{NH}_3\text{-N}$ concentration were not affected with decreasing dietary CP and RDP levels. Apparent N efficiency increased, and RDP N intake and predicted urine N output decreased, with decreased concentration of dietary CP and RDP in the diets ($P < 0.01$). Blood metabolites were not affected by treatments. In conclusion, to improve the efficiency of N utilization by early lactation dairy cow, 9.3% RDP in rations provides adequate protein to optimize milk production while minimizing N excretion in urine when the

amounts of lysine, methionine, and lysine to methionine ratio are balanced with sufficient dietary RUP.

Key words: nitrogen, rumen-degradable protein, dairy cows

INTRODUCTION

Accurate determination of animal protein requirements is critical for maximizing production and minimizing N excretion from dairy production (Huhtanen and Hristov, 2009). Interest is growing in developing practical nutritional strategies to reduce dietary CP levels without compromising cow's performance (Arriola-Apelo et al., 2014). A meta-analysis by Huhtanen and Hristov (2009) demonstrated the negative effect of increasing CP intake on N utilization efficiency in cows. Huhtanen et al. (2008) reported that the most effective strategy to improve milk N efficiency is to avoid feeding diets with excessively high CP concentration and especially excess RDP. Factors affecting dietary N utilization are complex and related to supplying sufficient amounts of RDP to meet the rumen N requirements, plus sufficient RUP of adequate intestinal digestibility (NRC, 2001). Inadequate RDP can lead to reduced ruminal $\text{NH}_3\text{-N}$ concentration, which causes a depression in DMI and microbial CP (Carlson et al., 2006). Excessive RDP will most likely be degraded to $\text{NH}_3\text{-N}$, absorbed into the blood, converted into urea in the liver, and excreted in urine as urea (Varel et al., 1999).

It appears from previous research that RDP recommended by NRC (2001) may be excessive for dairy cows (Armentano, 2007). The optimum concentration of RUP (5.5% of DM) fed in the current study was chosen based on our previous study (Bahrami-Yekdangi et al., 2014), in which decreasing CP and RUP to 16.4 and 5.5% of DM, respectively, increased DMI and had no negative effects on milk production. We hypothesized that decreasing RDP in diets could

maintain milk yield and improve N utilization in dairy cows without compromising ruminal digestion. Therefore, the aim of the study was to examine the effects of reduced dietary CP and RDP levels on production performance, nutrient digestibility, N utilization, and blood metabolite in dairy cows when dietary RUP is kept constant and the lysine to methionine ratio is balanced.

MATERIALS AND METHODS

Material and methods

Cows, Management, and Treatments

The study was conducted at the F.K.A. Agriculture and Animal Husbandry, Isfahan, Iran. Animals were cared for according to the guidelines of the Iranian Council of Animal Care (1995). Nine multiparous lactating Holstein cows (2nd parities, 585 ± 36 kg of BW; mean ± SD) averaging 50 ± 12 DIM and producing 48 ± 5 kg/d milk were used in the study. Cows were randomly assigned to a triplicate 3 × 3 Latin square with 3 dietary treatments. The dietary treatments (TMR) were: 1) a TMR containing 16.4% CP (10.9% RDP based on DM), 2) a TMR containing 15.6% CP (10% RDP based on DM), and 3) a TMR containing 14.8% CP (9.3% RDP based on DM). The daily CP requirements of Large Breed Cows (live weight of 680 kg at 90 d in milk) were 16.3 % for 45 kg/d with minimal changes in BW as predicted by NRC (2001) model. Chemical composition of the feed ingredients used in this experiment is given in Table 1. Rations were formulated by using the Cornell Net Carbohydrate and Protein System (CNCPS) model (Fox et al., 2004) and the feed analysis results (Table 1). Proximate analysis (CP, NDF, ADF, ash, ether extract, and NFC), degradability of protein (% of CP) by Cornell Net Carbohydrate and Protein System (CNCPS) method (Sniffen et al., 1992; Licitra et al., 1996), amino acid profile were determined using an automated Amino Acid analyzer (JLC-500/V,

JEOL, Tokyo, Japan) for feed ingredients. Chemical composition and lysine, methionine, and lysine to methionine ratios, predicted values of MP in experimental diets are reported in Table 2. Ruminally protected methionine was added to the diets to meet cow requirements (59 grams per d) for methionine using AC (AminoCow, the Mepron Ration Evaluator, version 3.5.2). All diets contained the same amounts of lysine (179.4 g of MP). There were three experimental periods. Each experimental period lasted 21 d, with 16 d for diet adaptation followed by 5 d for data collection. Cows were housed in tie-stalls and allowed to exercise for 1 h every afternoon. Total mixed ration was offered in equal amounts twice daily (0800 and 1600 h) allowing for *ad libitum* intake (5 to 10% orts) and fresh water. Feed efficiency was calculated as milk yield/DMI. Body condition score (Edmonson et al., 1989) was assessed on a scale of 1–5 (1 = emaciated, 5 = extremely obese) in increments of 0.25 at d 21 of each period by the same operator. Cows were milked 3 times per d at 0600, 1400, and 2200 h and milk production was recorded at each milking.

Sampling

On the last 5 consecutive days of each period, DMI was determined for each cow as the difference between TMR offered and orts weighed daily, with samples collected daily for the determination of DM and nutrient content. Additionally, feces were collected into large steel trays which were positioned over the gutter behind each stall on the last 5 consecutive days of each period. Feces were mixed thoroughly for each cow and 5% of daily output was sampled and stored at –20°C for later chemical analysis.

Milk samples were collected each day during d 16 to 21 of study from each cow from the 3 consecutive milkings. Milk samples were pooled in proportion to the corresponding milk yield and kept at room temperature (i.e., 23°C) with the preservative potassium dichromate. Fat-

corrected milk (FCM) yield was calculated as $(0.4 \times \text{kg of milk}) + 15 (\text{kg of milk} \times \text{milk fat}/100)$. Energy-corrected milk (ECM) yield was calculated as $(0.327 \times \text{kg of milk}) + (12.95 \times \text{kg of fat}) + (7.2 \times \text{kg of protein})$. Predicted urine N output (g/d) was calculated as $[0.0283 \times \text{milk urea N (mg/dl)} \times \text{BW (kg)}]$ (Wattiaux and Karg, 2004). Predicted fecal N (g/d) was calculated as $[\text{N intake} - \text{Predicted urine N} - \text{milk N}]$. In addition, apparent N efficiency was calculated as $[100 \times \text{milk N (g/d)} / \text{N intake (g/d)}]$. Cows were weighed at the start of the trial and on d 21 of each period.

On d 19 of each period, rumen fluid samples were collected 4 h after morning feeding using a stomach tube attached to a vacuum pump and strained through 4 layers of cheesecloth. Ruminal pH was immediately determined using a handheld pH meter (HI 8314 membrane pH meter, Hanna Instruments, Villafranca, Italy). Then, 20 ml of the rumen fluid was acidified with 0.4 ml of 50% (vol/vol) sulfuric acid and stored at -20°C for later $\text{NH}_3\text{-N}$ analysis (Broderick et al., 2004).

On d 20 of each period, blood samples were collected approximately 4 h after the morning feed from the coccygeal vessels using sterile tubes containing EDTA solution (Vacutainer, Becton Dickinson, Franklin Lakes, NJ, USA) and immediately placed on ice before further processing. Blood samples were centrifuged at $3,000 \times g$ for 15 min. Plasma samples were separated and stored at -20°C until analysis.

Chemical Analyses

Total mixed ration, feed ingredients, orts and fecal (pooled by cow within the period) samples were dried at 60°C for 48 h, ground through a 1-mm screen Wiley mill (standard model 4; Arthur M. Thomas, Philadelphia, PA, US). Samples were analyzed for DM (AOAC 2000;

method 930.15), ash (AOAC 2000; method 942.05), CP (AOAC 2000; method 990.06), and ether extract (method 920.39; AOAC 2000). Concentrations of ADF and NDF inclusive of residual ash were determined without sodium sulfide and with the inclusion of heat-stable α -amylase (100mL/0.5 g of sample) (Van Soest et al., 1991). The nonstructural carbohydrate content (% of DM) of the diets was calculated as $100 - (\text{NDF} + \text{CP} + \text{ether extract} + \text{ash})$. Organic matter was determined by ashing at 550 °C overnight. Acid insoluble ash was used as an internal marker to estimate the apparent total tract digestibility of DM, OM, ether extract, CP, NDF, and ADF (Van Keulen and Young, 1977). Apparent digestibility was calculated based on the relative concentrations of these nutrients and of AIA in the feed and feces.

Milk samples were analyzed for fat, total protein, and lactose, using a Milk-o-Scan analyzer (Foss Electric A/S, Hillerød, Denmark; AOAC, 1996). Milk urea nitrogen was measured automatically by the conduct metric-enzymatic method (CL10 micro analyzer, Eurochem, Roma, Italy; Cattani et al., 2014). Blood metabolite concentrations were spectrophotometrically (UNICCO, 2100, Zistchemi Co., Tehran, Iran) determined using commercially available kits (Pars Azmoon Company, Catalogue No.: glucose (1-500-017), triglyceride (1-500-032), total protein (1-500-028), albumin (1-500-001), blood urea N (BUN; 1-400-029), and aspartate aminotransferase (AST; 1-400-019), Tehran, Iran) according to the manufacturer's instructions. Plasma concentrations of non-esterified fatty acids (NEFA) were determined using a commercial assay kit (Wako Chemical USA, Richmond, VA, USA; Custer et al., 1983). Plasma β -hydroxybutyrate concentrations were analyzed using a commercial assay kit (BHBA kit 310-A UV; Sigma Chemical, St. Louis, MO, USA). Globulin concentrations were obtained by deducting albumin from total protein.

Statistical analysis

All the data were analyzed in a 3×3 Latin square design using PROC MIXED of SAS (SAS Institute Inc.). The model included square, period, cow (within square), treatment (effect of diet), square × treatment and period × treatment interactions, as well as overall error. All the variables were considered to be fixed, except the cow (within square) and overall error, which were taken to be random. The MIXED model was used for data analysis.

$$Y_{ijkl} = \mu + T_i + A_j + P_k + (T \times P)_{ik} + b_1 (P_{milk}) + e_{ijkl}$$

where, μ = overall mean, y_{ijke} =each observation, T_i = fixed effect of treatment, A_j = random effect of cow, P_k = fixed effect of period, $(T \times P)_{ie}$ = interaction of period and treatment, b_1 = covariate factor of initial milk production, and e_{ijkl} = random error term. Orthogonal polynomial contrasts (linear and quadratic) were used to examine treatment effects on the response variables. Effects of the factors were declared significant at $P \leq 0.05$ unless otherwise noted and trends were discussed at $P \leq 0.10$.

RESULTS

Feed intake, Milk yield and Composition

The data of DMI, feed efficiency, BW, milk urea N, milk yield and milk composition are presented in Table 3. Dry matter intake was not significantly affected by decreasing dietary CP and RDP levels. Milk production and composition, 4%FCM, ECM, Milk/DMI, FCM/DMI, and ECM/DMI were not affected by the decreasing levels of CP and RDP levels the diets. In spite of the linear decline in CP (from 16.4 to 14.8%) and RDP (from 10.9 to 9.3%) neither yield nor the percentage of milk composition (fat, protein, and lactose) were significantly affected by decreasing dietary CP and RDP levels. Milk urea N decreased linearly ($P < 0.05$) from 16.5 to

14.9 mg/dL with decreasing concentrations of CP and RDP in the diets. Body weight was not significantly affected by decreasing dietary CP and RDP levels.

Nitrogen Efficiency, Digestibility, and Rumen Parameters

The data of N efficiency, apparent digestibility of nutrients, and rumen parameters are presented in Table 4. Daily N intake linearly decreased ($P < 0.01$) with decreasing dietary CP and RDP levels, whereas the fecal N excretion did not change. Apparent N efficiency increased, and RDP N intake and predicted urine N output decreased, with decreased concentration of dietary CP and RDP in the diets ($P < 0.01$). In the current study, the lowest RDP diet (9.3 % DM) had most apparent N efficiency (31.98 %). In the current experiment, N efficiency increased from 29.32 to 31.98 % as RDP decreased from 10.9% to 9.3% of DM. Assuming that protein retention is negligible compared with protein used for milk production, N excretion in urine and feces can be calculated as shown in Table 4. Apparent digestibility of DM, CP, OM, ether extract, NDF, ADF, and ruminal pH and $\text{NH}_3\text{-N}$ concentration were not affected by decreasing dietary CP and RDP levels.

Blood Metabolites

The data of blood metabolites are presented in Table 5. Blood concentrations of glucose, triglyceride, BHBA, NEFA, AST, total protein, albumin, globulin, and albumin/globulin ratio were not significantly affected by decreasing dietary CP and RDP levels. However, a trend for a linear decline ($P = 0.06$) in BUN concentration was observed with decreasing concentrations of CP and RDP in the diets.

DISCUSSION

In the current study, RUP levels were constant across the treatments (5.5% of DM), whereas CP and RDP were gradually decreased and all diets were calculated to supply a post-rationally lysine to the methionine ratio of about 3:1 (Clark et al., 1992). We showed in our previous study (Bahrami-Yekdangi et al., 2014) that the NRC (2001) model predicts values for CP, RUP, and MP greater than requirements of early lactating dairy cows. Reduction in dietary CP (from 18 to 15.6% of DM), RUP (from 7.1 to 4.7% of DM), and MP (from 2,867 to 2,366 g/d) in the diets of early lactating dairy cows had no significant effects on milk production and composition. Furthermore, decreasing dietary CP (from 18 to 15.6% of DM) increased N efficiency without any detrimental effect on performance of the animal during early lactation. It is well known that an imbalance of AA available to the dairy cows will impair milk and milk protein production. Methionine and lysine are identified as the most limiting AA in lactating dairy cows (Schwab et al., 2007). Balancing diets for AA provides the opportunity to supply similar amounts of the most limiting AA with reduced or similar concentrations of RUP (Schwab et al., 2007). The NRC (2001) recommends a concentration of 7.2 and 2.4%, respectively, for lysine and methionine in MP for maximal use of MP for milk protein production. In the present study, the predicted ruminal microbial supply did not decrease with decreasing RDP level, because RDP decreased only slightly from 10.9 to 9.3. These findings are consistent with those of other studies that reported no changes in production performance when dietary CP varied from 16.7 to 18.4% (Davidson et al., 2003), from 16.4 to 20.4% (Mulligan et al., 2004), from 16.4 to 18.0% (Wattiaux and Karg, 2004), and from 14.6 to 18.3% (Castillo et al., 2001).

Feed intake, Milk yield, and Composition

Previous research has indicated that inadequate RDP (7.4% DM) can lead to reduced ruminal $\text{NH}_3\text{-N}$ concentrations and DMI (Gressley and Armentano, 2007) may be reduced when insufficient quantities of $\text{NH}_3\text{-N}$ are provided for microbial fermentation (Faverdin, 1999). However, our results herein indicated that 9.3% RDP diets appeared to be sufficient for N required by the rumen microbes and the level of RDP did not affect the DMI by early lactation dairy cow. Previously, Gressley and Armentano (2007) found no changes in DMI when a 7.4% RDP diet was fed as compared with a 10.4% RDP diet to lactating dairy cows. Cyriac et al. (2008) indicated that the DMI of cows fed low RDP diets (7.6% of dietary DM) were significantly lower than cows receiving greater dietary levels of RDP (8.8, 10.1, and 11.3% of dietary DM), but no differences in DMI of cows fed greater levels of RDP were observed.

It is well known that, as long as carbohydrates are not limiting, bacterial N and bacterial efficiency continue to increase as RDP increases in diets (Stokes et al., 1991). In the current study, cows fed the lowest RDP diets produced the same amount of milk, 4%FCM and ECM as cows fed the highest RDP diets. In a production study with lactating dairy cows (Armentano et al., 1993), it was found that increasing RDP from 9.5 to 11.7% did not result in increased milk production, indicating that there was no benefit in using diets with RDP levels greater than 9.5%.

Our findings are consistent with those of other studies that reported no changes in milk composition when dietary CP decreased from 19.4% to 16.8 (Davidson et al., 2003), from 18.0 to 16.4 % (Wattiaux and Karg, 2004), and from 16.1% to 13.5 (Gressley and Armentano, 2007), indicating that RDP at 9.3% DM was sufficient to support milk yield and its composition by early lactation dairy cow. Previous studies (Davidson et al., 2003; Reynal and Broderick, 2005; Kalscheur et al., 2006) have reported that the linear decrease in dietary CP level causes a subsequent decrease in milk urea concentration, which can be explained by decreasing ruminal

NH₃-N concentration. Our results are consistent with the report of Davidson et al. (2003), who observed no changes in milk urea N when dietary CP level varied from 19.4% to 16.8 DM (RDP = 7.4 % of DM) by early lactation dairy cow.

Nitrogen Efficiency, Digestibility, and Rumen Parameters

Fecal N consists primarily of indigestible microbial protein produced in the rumen as well as endogenous proteins, sloughed cells from the gastrointestinal tract, and undigested feed proteins (Mason, 1969; Davidson et al., 2003). The NRC (2001) estimates that 85% of RDP can be converted to microbial CP when RDP is the limiting factor to microbial growth. The metabolizable microbial CP is 64%, which is used for lactation at an efficiency level of 67% after accounting for maintenance, lactation, and growth. This leads to an estimated overall coefficient of 36%, which is greater than the 29-31% levels observed in this study; however, some cows may have been fed RDP in excess of their requirements. The results of this study suggest a linear response to RDP deficiency at a lower magnitude than that predicted by NRC (2001) model.

Maximizing the synthesis of microbial protein as a relatively inexpensive source of readily digestible protein in the small intestine is desirable; however, inefficiency of protein use within the animal increases as RDP increases, causing concerns of increased N excreted as waste. In agreement with Kalscheur et al. (2000) and Hristov et al. (2004), the predicted value of urinary N excretion increased linearly as RDP increased in the diet of dairy cows. Broderick et al. (2008) noted that total-tract digestibility of OM, NDF, and ADF is related to ruminal NH₃-N and dietary CP concentration. Insufficient RDP could lead to a ruminal NH₃-N deficiency that would depress microbial growth and fiber digestion (Allen, 2000). However, the lowest ruminal NH₃-N concentrations measured in the current study were still above the minimum

concentrations required (≥ 5 mg/dl) for rumen microbial growth (Satter and Roffler, 1975). NH_3 -N averaged 8.90 and 8.05 mg/dL in the rumen in this trial when diets contained, respectively, 16.4 and 14.8% CP.

Blood Metabolites

In the current study, the trend of a linear decline in BUN agreed with the significant decrease in milk urea N and the numerical decrease in rumen NH_3 -N concentration. This was expected because rumen NH_3 -N not incorporated into microbial protein is absorbed across the rumen wall and converted to urea in the liver for either excretion in urine, secretion in milk, or recycling to the rumen through saliva (Davidson et al., 2003); however, these differences were not statistically significant.

CONCLUSIONS

In conclusion, the present study confirm that reducing dietary CP (from 16.4 to 14.8% of DM) and RDP (from 10.9 to 9.3 % of DM) with the constant concentration of RUP (5.5% of DM) in high producing dairy cow diets, formulated to contain a post-rationally available Lys to the Met ratio of approximately 3:1, does not adversely affect milk production or composition. Reducing dietary CP and RDP levels in dairy cow diets increased the efficiency of N capture and decreased the amount of urinary N losses. In addition, milk urea N concentrations decreased linearly from 16.5 to 14.9 mg/d as RDP decreased in the diet of dairy cows, when correct dietary RUP, lysine, and methionine levels are included. Thus, in order to improve the efficiency of N utilization by early lactation dairy cow, 9.3% RDP in rations provides adequate protein to optimize milk production while minimizing N excretion.

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399 **Table 1.** Chemical composition of feed ingredients based on DM¹ (%) and for RUP and RDP based on CP (%) expressed as means

Ingredients	Composition, % of DM											
	DM	CP	RDP	RUP	NDF	ADF	Fat	Ash	NFC	Lysine	Methionine	
Alfalfa hay	93.2	13.5	73.7	26.2	55.7	38.8	3.4	8.5	18.7	0.19	0.73	
Corn silage	22.5	10.4	84.5	15.4	52.0	31.0	5.2	8.0	24.3	0.14	0.22	
Beet pulp	96.8	10.3	73.9	26.0	48.0	22.0	1.0	5.6	34.9	0.12	0.48	
Ground barley	89.5	10.7	60.4	39.5	22.5	5.0	2.0	4.0	60.7	0.16	0.37	
Ground corn	89.8	8.4	60.7	39.2	11.0	5.2	3.9	5.5	71.2	0.16	0.25	
Soybean meal mechanical pressure	92.0	47.4	62.8	37.2	18.0	9.3	3.0	6.5	25.0	0.61	2.71	
Extrude soy bean (full fat)	94.6	38.0	50.6	49.3	18.0	14.0	18.7	9.3	15.9	0.48	1.03	
Canola meal	92.8	37.3	69.4	30.5	33.5	18.7	3.3	8.2	17.6	0.68	1.94	
Cottonseed whole with lint	94.4	19.3	70.0	29.9	52.9	37.0	19.3	3.9	4.5	0.26	0.79	
Corn gluten meal	93.9	56.9	40.0	59.9	13.8	7.2	1.8	5.0	22.3	1.42	0.79	
Fish meal	95.6	60.1	34.2	65.8	6.5	8.5	17.7	15.0	0.6	1.61	1.82	

400 ¹Proximate analysis (CP, NDF, ADF, ash, ether extract, and NFC), degradability of protein (% of CP) by Cornell Net Carbohydrate
 401 and Protein System (CNCPS) method (Sniffen et al., 1992; Licitra et al., 1996), AA profile were determined using an automated
 402 Amino Acid analyzer (JLC-500/V, JEOL, Tokyo, Japan) for feed ingredients.

Table 2. Ingredients and chemical compositions (% of DM, unless otherwise noted) of the experimental diets.

Item	Dietary CP, ¹ % of DM		
	16.4	15.6	14.8
Ingredients (% of DM)			
Alfalfa hay	16.3	16.3	16.3
Corn silage	21.4	21.4	21.4
Beet pulp	3.8	3.8	3.8
Ground barely	17.7	17.7	17.7
Ground corn	17.7	18.9	20.2
Soybean meal (45% CP)	6.2	5.2	4.6
Soy bean extrude	5.5	5.5	5.5
Canola meal	1.1	1.1	1.1
Cottonseed whole	3.3	3.3	3.3
Corn gluten meal	1.8	1.8	1.8
Fish meal	0.8	0.8	0.8
Energy Booster ²	1.5	1.5	1.5
Sodium bicarbonate	0.8	0.8	0.8
Magnesium oxide	0.2	0.2	0.2
Calcium carbonate	0.6	0.6	0.6
Di-calcium phosphate	0.2	0.2	0.2
White Salt	0.3	0.3	0.3
Mineral and vitamin premix ³	0.4	0.4	0.4
Mepron ⁴	0.05	0.05	0.05
Urea 281% CP	0.22	0.12	0.01
Chemical composition			
NE _i (Mcal/kg DM)	1.74	1.74	1.74
CP (% of DM)	16.4	15.6	14.8
RDP (% of DM)	10.9	10.1	9.3
RUP (% of DM)	5.5	5.5	5.5
MP (g/d) ⁵	2533	2537	2536
MP from bacteria (g/d) ⁵	1138	1149	1155
MP from RUP (g/d) ⁵	1394	1388	1381
Lysine (g of MP) ⁵	179.4	179.4	179.4
Methionine (g of MP) ⁵	59.3	59.4	59.3
Lysine (% of MP) ⁵	7.1	7.1	7.1
Methionine (% of MP) ⁵	2.3	2.3	2.3
Lysine/Methionine ⁵	3.0	3.0	3.0
Ether extract (% of DM)	6.2	6.2	6.2
NDF (% of DM)	32.7	33.6	34.2
ADF (% of DM)	18.6	18.9	19.1
NFC ⁶ (% of DM)	38.3	38.6	38.9
Calcium (% of DM)	0.81	0.82	0.82
Phosphorus (% of DM)	0.43	0.43	0.43

¹ Treatments 1 to 3 consisted of total mixed ration containing 16.4%, 15.6%, and 14.8% CP, respectively.

² Energy Booster 100 Advance (MS Specialty Nutrition, Dundee, IL).

³ Composition: 195 g/kg of Ca, 21 g/kg of Mg, 2.2 g/kg of Mn, 0.3 g/kg of Zn, 0.3 g/kg of Cu, 0.12 g/kg of I, 0.1 g/kg of Co, 600.000 IU/kg of vitamin A, 200.000 IU/kg of vitamin D, and 0.2 g/kg of vitamin E, 2.5 g/kg antioxidant.

⁴ Rumen-protected Met product from Degussa Corp., (Kennesaw, Georgia, USA).

⁵ Estimated using the NRC (2001) model.

⁶ $NFC = [100 - (\%NDF - NDIN \times 6.25) - \%CP - \%Fat - \%Ash]$. Where NDIN = neutral detergent-insoluble CP.

Table 3. Effects of decreasing dietary CP (from RDP sources) on DMI, feed efficiency, BW, milk urea nitrogen, milk yield and composition

Item	Dietary CP, ¹ % of DM			SEM	P-value	
	16.4	15.6	14.8		linear	quadratic
DMI (kg/d)	26.28	26.74	26.03	0.44	0.24	0.49
Milk production (kg/d)	41.64	41.00	40.61	1.30	0.74	0.16
4% FCM ² (kg/d)	35.26	36.16	35.04	1.65	0.26	0.27
ECM ³ (kg/d)	38.52	39.17	38.13	1.60	0.61	0.76
Milk/DMI ⁴	1.58	1.53	1.55	0.04	0.49	0.55
4% FCM/DMI ⁵	1.30	1.30	1.33	0.05	0.67	0.39
ECM/DMI ⁶	1.42	1.42	1.45	0.05	0.81	0.25
Milk composition						
Fat (%)	2.97	3.22	3.06	0.15	0.19	0.31
Protein (%)	3.04	3.02	3.03	0.09	0.18	0.09
Lactose (%)	4.98	4.97	4.99	0.05	0.13	0.16
Fat (kg/d)	1.24	1.32	1.25	0.09	0.81	0.25
Protein (kg/d)	1.26	1.24	1.23	0.03	0.33	0.50
Lactose (kg/d)	4.66	4.52	4.62	0.04	0.17	0.48
Milk urea N (mg/dl)	16.5	15.2	14.9	0.47	0.05	0.11
Mean BW (kg)	583.2	578.4	592.5	10.11	0.77	0.11

¹ Treatments 1 to 3 consisted of TMR containing 16.4%, 15.6%, and 14.8% CP, respectively.

² Fat-corrected milk (kg/d) = (0.4 × kg of milk) + 15 (kg of milk × milk fat/100).

³ Energy-corrected milk (kg/d) = milk yield × 0.327 + milk protein yield × 7.2 + milk fat yield × 12.95.

⁴ Milk/DMI (feed efficiency ratio) = milk production (kg/d) : DMI (kg/d).

⁵ FCM/DMI = FCM production (kg/d) : DMI (kg/d).

⁶ ECM/DMI = ECM production (kg/d) : DMI (kg/d).

Table 4. Effects of decreasing dietary CP (from RDP sources) on nitrogen (N) retention, N efficiency, apparent digestibility of nutrients, and rumen parameters

Item	Dietary CP ¹ , % of DM			SEM	P-value	
	16.4	15.6	14.8		linear	quadratic
Intake N (g/d)	689.5	664.5	618.8	11.11	<0.01	0.19
Intake RDP N (g/d)	454.0	427.9	383.5	7.17	<0.01	0.27
Intake RUP N (g/d)	235.5	236.6	235.3	7.17	0.89	0.55
Milk N (g/d)	202.3	199.5	197.7	5.50	0.45	0.33
Predicted urine N (g/d) ²	341.0	348.1	281.0	7.26	0.03	0.11
Predicted fecal N (g/d) ³	143.8	140	138	6.33	0.23	0.28
Apparent N efficiency (%) ⁴	29.32	29.88	31.98	0.68	<0.01	0.41
Total tract apparent digestibility (%)						
DM	73.63	74.33	74.28	3.61	0.50	0.22
CP	68.97	69.33	69.21	3.10	0.82	0.32
OM	76.24	77.28	78.34	2.98	0.34	0.79
Ether extract	68.54	67.43	68.32	3.23	0.30	0.56
NDF	45.89	47.34	46.99	2.94	0.37	0.45
ADF	42.45	43.64	43.51	3.31	0.39	0.33
Rumen parameters						
Rumen pH	6.1	6.0	5.9	0.31	0.26	0.63
Rumen NH ₃ -N (mg/dl)	8.90	8.11	8.05	0.37	0.43	0.55

¹ Treatments 1 to 3 consisted of TMR containing 16.4%, 15.6%, and 14.8% CP, respectively.

² Predicted urine N output = 0.0283 × milk urea N (mg/dl) × BW (kg), (Wattiaux and Karg, 2004).

³ Predicted fecal N (g/d) = N intake – Predicted urine N – milk N.

⁴ Apparent N efficiency (g/d) = 100 × milk N (g/d) / N intake (g/d).

Table 5. Effects of decreasing dietary CP (from RDP sources) on blood metabolites

Item ²	Dietary CP, ¹ % of DM			SEM	P-value	
	16.4	15.6	14.8		linear	quadratic
Glucose (mg/dl)	58.7	54.3	58.2	3.88	0.55	0.57
Triglyceride (mg/dl)	18.8	17.9	19.4	2.87	0.74	0.70
BHBA (mEq/l)	0.2	0.3	0.3	0.31	0.54	0.54
NEFA (mEq/l)	0.3	0.4	0.3	0.35	0.33	0.65
BUN (mg/dl)	17.5	15.6	15.1	2.31	0.06	0.53
AST (U/l)	81.3	79.4	72.5	4.41	0.54	0.87
Total protein (g/dl)	7.9	8.3	8.5	0.54	0.43	0.49
Albumin (g/dl)	3.6	3.7	3.6	0.45	0.65	0.42
Globulin (g/dl)	4.6	5.0	4.6	0.42	0.53	0.42
Albumin: Globulin	0.65	0.70	0.64	0.10	0.19	0.65

¹ Treatments 1 to 3 consisted of TMR containing 16.4%, 15.6%, and 14.8% CP, respectively.

² NEFA = non-esterified fatty acids; BHBA = β -hydroxy butyric acid; BUN = plasma urea N; AST = Aspartate transaminase.